OneRhythm "Shared Rhythm Distance" executive brief

Every 10-second ECG recording stretches 25 centimeters on paper. If the world's 52.55 million atrial fibrillation patients each contributed a single strip, their combined rhythms would stretch 13,138 kilometers — one-third of the way around the Earth. That single statistic contains the entire emotional logic of the Shared Rhythm Distance feature: individual heartbeats, invisible and isolating when experienced alone, become a collective journey visible to everyone when combined. This brief synthesizes verified epidemiological data, ECG standardization math, privacy-safe generative art techniques, behavioral psychology, and implementation architecture into an authoritative reference for building this feature.

The clinical urgency is real. The EHRA's March 2025 declaration that **1 in 3 people will develop a potentially life-threatening heart rhythm disorder** in their lifetime (ESC) (EurekAlert!) means the addressable community is not a niche — it is 2.73 billion people. The UK Biobank's 2025 JAHA study confirms that loneliness and social isolation are independent risk factors for AF progression, (AHA Journals) with persistent loneliness doubling AF risk. (ScienceDirect) A feature that transforms anonymous ECG contributions into shared art and measurable collective distance addresses a documented, modifiable cardiovascular risk factor: isolation itself.

The global arrhythmia burden is massive and accelerating

The most current authoritative estimate comes from the Global Burden of Disease 2021 study, published in *EP Europace* (2024). **Atrial fibrillation and atrial flutter affect 52.55 million people worldwide**, with 4.48 million new cases and 340,000 attributable deaths annually. (ScienceDirect) This represents a **137% increase** in absolute cases since 1990, though the age-standardized prevalence rate has remained relatively stable at ~621 per 100,000, (PubMed Central) indicating that population aging and growth — not increasing individual risk — drive the surge. The earlier GBD 2019 estimate of 59.7 million reflects different modeling methodology rather than an actual decline. (BMC Public Health)

AF is the dominant arrhythmia tracked by GBD, but it represents only a fraction of total rhythm disorders. Paroxysmal supraventricular tachycardia affects an estimated **18 million people** globally (extrapolated from a prevalence of 2.25 per 1,000). (ScienceDirect) Wolff-Parkinson-White pattern appears in 2–4 per 1,000 of the general population, mostly asymptomatic. (AHA Journals) ARVC affects 1 in 2,000–5,000; (NCBI) Long QT syndrome 1 in 2,000–7,000; Brugada syndrome peaks in Southeast Asia at up to 1 in 1,000. (PubMed Central) Adding bradyarrhythmias requiring pacing (~5–10 million symptomatic), the total global population with any clinically identifiable cardiac arrhythmia likely exceeds **100 million**, though no single aggregate estimate exists because GBD only systematically tracks AF/AFL. (PubMed Central) An estimated **1–2 million people**

currently live with implanted cardiac defibrillators, and approximately 4–5 million live with pacemakers.

Regional variation is substantial and relevant to the "countries represented" counter. The United States has 3–6 million current AF cases, projected to reach (AHA Journals) **12.1 million by 2030**. (Frontiers) Europe carries approximately 9 million cases, projected to reach 14–17.9 million by 2060. (AER Journal) East Asia reports the highest absolute incidence among young adults. Africa, paradoxically, has the lowest current detection rates but may exceed the US, China, or India in AF population by 2050 due to demographic transitions and improved detection. (AER Journal) The epidemiological pattern varies up to **12-fold between regions**, (PubMed Central) driven by aging demographics, obesity rates, diagnostic access, and genetic factors.

Verifying the "1 in 3" lifetime risk claim

The EHRA's assertion was published March 1, 2025, as part of Pulse Day, authored by Professor Julian Chun (EHRA Scientific Initiatives Committee Chair) and Professor Helmut Pürerfellner (EHRA President). (ESC) It draws primarily on Linz et al. (2024) in *Lancet Regional Health – Europe*, which states lifetime risk of AF is "about 1 in 3–5 individuals after the age of 45 years," (The Lancet) and on the Framingham Heart Study, which found a ~1 in 3 lifetime risk in white individuals. (AHA Journals) The claim is best characterized as an evidence-based advocacy statement — robust for AF in developed-country cohorts, extended to all heart rhythm disorders as a generalization. For OneRhythm's purposes, it is appropriate to cite with attribution: "According to the European Heart Rhythm Association, 1 in 3 people worldwide will develop a potentially life-threatening heart rhythm disorder in their lifetime."

The math of rhythm distance, from strip to circumnavigation

Standard ECG strip dimensions

The universal clinical standard, defined by AHA/ACCF/HRS (2007), specifies a resting 12-lead ECG recorded at **25 mm/s for 10 seconds**, producing a strip exactly **250 mm (25 cm)** long. This is confirmed across ACC/AHA Clinical Competence Statements, BMJ educational materials, and every major cardiology textbook. The 25 mm/s speed means each small box (1 mm) represents 40 ms and each large box (5 mm) represents 200 ms. (EcgWaves) (PubMed Central) A voltage calibration of 10 mm/mV is standard. (AHA Journals) (My-ekg)

Variations exist but do not displace this canonical standard. Pediatric and tachyarrhythmia ECGs sometimes use **50 mm/s** (EcgWaves) (PubMed Central) (producing a 500 mm strip for the same 10 seconds), and rhythm strips can extend to 30 seconds or longer. Critically, **all major consumer wearable ECGs converge on a 30-second recording**: Apple Watch (30 seconds, fixed), (Apple Support) Samsung Galaxy Watch (30 seconds, fixed), (Samsung) Withings ScanWatch (30 seconds, fixed), (Withings) and AliveCor KardiaMobile (30 seconds default, adjustable up to 5

minutes). AliveCor KardiaCare At 25 mm/s, a 30-second wearable recording produces a strip of **750 mm (75 cm)**.

Core calculations

All calculations use Earth's equatorial circumference of **40,075.017 km** VCalc Wikipedia (WGS84 geodetic standard, confirmed by NASA).

How many standard 10-second ECG contributions circle the Earth once?

Parameter	Value
One standard strip length	0.25 m (25 cm)
Earth circumference	40,075,017 m
Strips required	**160,300,068** (~160.3 million)

If all 52.55 million AF/AFL patients contributed one standard strip:

Parameter	Value
Total distance	52,550,000 × 0.25 m = **13,137.5 km**
Percentage of Earth circumference	**32.8%**
Narrative equivalent	Tampa to Tokyo, or more than the entire width of the Atlantic Ocean

If the lifetime-risk cohort (1 in 3 of ~8.2 billion) each contributed one strip:

Parameter	Value
Cohort size	~2,733,333,333
Total distance	2,733,333,333 × 0.25 m = **683,333 km**
Times around Earth	**17.05 times**
Narrative equivalent	Nearly twice the distance from Earth to the Moon (384,400 km)

What fraction of the lifetime-risk cohort completes one loop?

Parameter	Value
Strips needed	160,300,068
As percentage of 2.73 billion	**5.87%**
Absolute number	~160.3 million people

The wearable-length insight

If OneRhythm accepts 30-second wearable ECG recordings (the consumer device standard), each contribution is 75 cm rather than 25 cm. This transforms the calculus dramatically:

Scenario (30-second strips)	Distance	Earth %
52.55M AF patients × 1 strip each	**39,412.5 km**	**98.3%**
~53.43 million contributions	40,075 km	**100% — one full loop**

This means the global AF/AFL population contributing a single wearable-length strip would nearly complete one full circumnavigation — **98.3% of the way around the Earth**. This is the feature's foundational narrative: *the AF community, together, can circle the planet.*

Privacy-safe transformation of ECG into generative art

ECG is a biometric identifier requiring serious protection

Raw ECG waveforms are not benign data. Research demonstrates **80–96% re-identification accuracy** from ECG morphology alone, even across sessions 6–18 months apart. A 2025 large-scale study (arXiv 2602.02776) achieved TAR=0.908 at FAR=1e-3. Even extracted PQRST features (without full waveforms) enable 82% participant re-identification. arXiv The Lancet Digital Health (2022) warned explicitly that "deidentification of data from wearable devices gives us a false sense of security." The Lancet Under HIPAA's Safe Harbor method, biometric identifiers — which explicitly include ECG waveforms — must be removed or transformed before data is considered de-identified. John Snow Labs

State of the art in ECG anonymization

The field offers several approaches, none yet providing a universally accepted framework:

GAN-based anonymization (PrivECG, MLHC 2023) regenerates 12-lead ECGs with modified R-wave morphology, reducing sex prediction accuracy from 0.876 to near-random 0.529 (Proceedings of Machine Le…) and increasing identification EER from 0.098 to 0.251 (Mhi-omics) — meaningful but incomplete anonymization. Disease prediction F1 score drops only from 0.931 to 0.885, demonstrating that clinical utility can be partially preserved. (Proceedings of Machine Le…) The REACT framework (2024) extends this with reinforcement learning-based adaptive anonymization.

Differential privacy has been applied through noise addition to raw signals (Laplace/Gaussian mechanisms), DP-SGD in federated ECG classification, and Rényi-differentially private GANs. (arXiv) The fundamental tension: ECG's narrow physiological operating range means noise that provides strong privacy guarantees often destroys clinical utility. The mSieve framework offers a novel "differential behavioral privacy" approach using data substitution rather than noise, achieving 90% inference accuracy while preserving behavioral privacy. (PubMed Central)

Synthetic ECG generation via mathematical modeling (McSharry et al., IEEE TBME 2003) and deep generative models can produce morphologically realistic signals that carry no biometric link to any individual. A 2024 scoping review identified 79 studies across mathematical, computer vision, and deep generative approaches.

Artistic transformation as the anonymization layer

For OneRhythm's specific use case, the artistic transformation itself can serve as de-identification — **if and only if** the transformation is sufficiently lossy, non-invertible, and destroys individual morphological signatures. The recommended approach:

1. **Extract only high-level, non-biometric features** from the ECG: average heart rate, rhythm regularity score (regular vs. irregular), rhythm density (beats per second), and waveform complexity (a derived metric). These aggregate features cannot reconstruct the original waveform's PQRST morphology.
2. **Map extracted features to artistic parameters** using one-way transformations: rhythm regularity → pattern symmetry; heart rate → color temperature; complexity → texture density; amplitude variance → opacity/glow intensity.
3. **Never store, transmit, or display the raw ECG waveform.** The waveform is processed locally on the device, features are extracted, and only the derived artistic parameters (not the signal) leave the device.
4. **Obtain Expert Determination** under HIPAA's de-identification framework from a qualified statistical expert certifying that the artistic output carries very small (<0.04–0.05 probability) re-identification risk.

Precedent projects demonstrate feasibility

Several projects have successfully transformed biometric data into generative art. Empatica's **Cocoon** installation (2022) merged pulse rate, electrodermal activity, breathing rate, and movement from 24 FDA-cleared smartwatches into a 12,000-LED circular sculpture where color and light patterns reflected collective physiological states. Empatica The **heart/work** iOS app generates unique, non-repeatable artworks from Apple Watch heartbeat data using a custom visual language. MPRECKE Medium Justus Harris's **MedSculp** project transforms continuous glucose monitor data into 3D-printed sculptures where blood sugar spikes become physical protrusions — compressing ~2,000 data points into abstract forms where individual readings are not recoverable. School of the Art Institute ... MedCentral Giorgia Lupi's **Data Humanism** framework, formalized in 13 principles, argues that data visualization should "convey knowledge AND inspire feelings simultaneously" while embracing imperfection and human context. giorgialupi

The common mapping pattern across these projects:

ECG Feature (Non-Biometric)	Artistic Parameter	Effect
Rhythm regularity	Pattern symmetry/flow	Regular rhythms → smooth gradients; irregular → complex, organic textures
Heart rate (BPM range)	Color temperature	Lower → cool blues/purples; higher → warm ambers/reds
Beat-to-beat variability	Texture density	Low variability → clean; high variability → rich, layered
Waveform complexity	Opacity and layering	Simple morphology → translucent; complex → deep, saturated
Recording duration	Tile size/density	Longer contributions → more detailed tiles

FDA regulatory position

An artistic ECG visualization that makes **no diagnostic claims** would not qualify as Software as a Medical Device (SaMD) under FDA's framework. The FDA's classification depends on intended use: Bipartisan Policy Center NAMSA if the software transforms ECG into art purely for aesthetic, wellness, or self-expression purposes, it falls outside SaMD scope. Critical safeguards: no language suggesting diagnostic capability, no claims about detecting or monitoring conditions, clear labeling that "this is not a medical device," and de-identification of ECG data before any artistic transformation. The FDA exercises enforcement discretion for low-risk wellness and self-management tools. Bipartisan Policy Center

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Why isolation reduction is both emotionally and clinically justified

Loneliness independently worsens arrhythmia outcomes

The UK Biobank–JAHA 2025 study (Qiao et al., *Journal of the American Heart Association*, 2025;14(9):e039885) followed **418,656 participants** without AF for a median of 14.7 years using a multistate model from baseline through incident AF to major adverse cardiovascular events (MACE) to death. Both loneliness and social isolation were independently associated with increased AF incidence and adverse progression. AHA Journals A supporting UK Biobank study (Li et al., *European Journal of Medical Research*, 2024) found loneliness conferred a hazard ratio of **1.11 (95% CI 1.07–1.16)** for incident AF across 441,056 participants over 13.9 years. Springer Most strikingly, a third UK Biobank analysis found **persistent loneliness doubled AF risk (HR 2.11, 95% CI 1.08–4.11)** and loneliness combined with anxiety increased risk by **130% (HR 2.30, 95% CI 1.59–3.33)**. ScienceDirect The AHA's 2022 Scientific Statement formally established social isolation and loneliness as risk factors for cardiovascular morbidity and mortality.

This evidence means a feature designed to reduce isolation among arrhythmia patients is not merely engagement design — it addresses a documented, modifiable cardiovascular risk factor.

The psychology of contributing to a shared artifact

Three interlocking psychological mechanisms make collective data art effective as a belonging intervention:

The **IKEA effect** (Norton, Mochon, Ariely, 2012) demonstrates that people value what they help create by **63% more** than identical pre-made objects. Wikipedia Crucially, collaborative IKEA effect studies show the bias is equally strong in group settings — feelings of togetherness and collective problem-solving are supported by shared labor. PubMed Central For OneRhythm, each contributed rhythm tile represents labor that generates psychological ownership of the collective mosaic.

The **goal-gradient effect** (Hull, 1932; Kivetz et al., 2006) shows people accelerate effort as they approach a goal. Laws of UX In charitable contexts (Cryder, Loewenstein, Seltman, 2013), donors contribute more when campaigns are closer to targets because they experience a **heightened sense of personal impact**. The Earth-circling milestone creates exactly this dynamic: as the counter approaches 40,075 km, each contribution feels increasingly consequential.

The **endowed progress effect** (Nunes & Dreze, 2006) found that loyalty cards starting with artificial head starts were redeemed **82% more often** than equivalent zero-start cards. Lead Alchemists OneRhythm can leverage this by launching the counter with seed contributions or presenting early milestones as achievable (e.g., crossing the English Channel at just 34 km).

Lessons from the AIDS Quilt and visible solidarity campaigns

The NAMES Project AIDS Memorial Quilt — 50,000+ panels, 110,000 names, Wikipedia 54 tons Crafts Council — remains the most powerful precedent for making collective suffering visible through participatory art. Its mechanisms are directly transferable: **individualization of loss** (each 3'×6' panel tells one story), the **domestic form as destigmatization** Villanova (quilting reframed AIDS from a "gay disease" to an "American disease"), Crafts Council and the **making process as intervention** ("many of these people have not had their names repeated since they passed, because so many of them died in a veil of shame"). National Endowment for th... The Quilt works because makers *choose* how to represent their loved ones — agency over narrative is essential.

The breast cancer pink ribbon campaign achieved measurable impact — mammography utilization rose from **29% to 70%** among women 40+ between 1987 and 2000 ScienceDirect — but also generated instructive backlash. "Pinkwashing" (companies using ribbons to sell products without supporting the cause), Breastcancer.org imposed identity labels ("warrior," "survivor" that alienated some patients), Breastcancer.org and trigger potential ("I already feel overwhelmed and triggered by Breast Cancer Awareness Month") Breastcancer.org all offer cautionary lessons. The core design principle: **make participation optional, empowering, and non-extractive.**

Implementation architecture and data model

Standardizing the Rhythm Unit

The **canonical Rhythm Unit should be 0.25 meters (25 cm)**, corresponding to one standard 10-second clinical ECG strip at 25 mm/s. This unit is universal, clinically grounded, and creates manageable math (160.3 million units per Earth circumference). For wearable ECG contributions (30-second recordings), one contribution equals **3 Rhythm Units** (75 cm). PubMed Central This normalization allows mixing clinical and consumer device contributions while maintaining a single consistent counter.

Source	Recording Length	Rhythm Units	Physical Distance
Standard 12-lead ECG	10 seconds	1	25 cm
Apple Watch / Samsung / Withings	30 seconds	3	75 cm
AliveCor KardiaMobile (default) KardiaCare	30 seconds	3	75 cm
AliveCor KardiaMobile (extended) AliveCor	Up to 5 minutes	Up to 30	Up to 7.5 m

Counter architecture

For a counter targeting millions of contributions, an **event-driven materialized counter** is the performance-optimal approach, modeled on Netflix's distributed counter abstraction (which handles 75,000 requests/second at single-digit millisecond latency). ByteByteGo

Recommended stack:

The write path should use **Redis with atomic INCR** for the hot counter, providing O(1) increment operations with sub-millisecond latency. System Design Each contribution event triggers an atomic increment of the global `rhythm_distance_cm` counter and per-country sub-counters. A write-behind mechanism periodically snapshots the counter to a durable store (PostgreSQL or DynamoDB) for crash recovery. System Design

The read path should serve the counter value from a **CDN-cached endpoint** refreshed every 30–60 seconds. Client-side polling at this interval provides near-real-time feel without WebSocket overhead. For milestone moments (when the counter crosses a threshold), Server-Sent Events (SSE) can push celebratory notifications to active sessions. System Design

Milestone detection should use a threshold table:

```
milestones (
  milestone_id    SERIAL PRIMARY KEY,
  name            TEXT,          -- "Across the English Channel"
  distance_cm     BIGINT,       -- 3,400,000 (34 km)
  reached_at      TIMESTAMPTZ,  -- NULL until achieved
  narrative       TEXT           -- "Our shared rhythm just crossed the English
Channel"
)
```

A background worker polls the counter every 10 seconds and fires milestone events when thresholds are crossed. Milestone events trigger push notifications, in-app celebrations, and social-sharing card generation.

Data model per contribution

What to store:

Field	Type	Purpose
`contribution_id`	UUID	Unique identifier
`timestamp`	TIMESTAMPTZ	When contributed
`rhythm_units`	INTEGER	Number of 25 cm units (1 for clinical, 3 for wearable)
`visibility`	ENUM (public/private)	Whether tile appears in community mosaic
`condition_category`	ENUM (optional)	Broad category: AF, SVT, bradycardia, other, prefer-not-to-say
`country_code`	ISO 3166-1	For "countries represented" counter
`art_tile_hash`	TEXT	Reference to generated tile in asset storage
`device_type`	ENUM	Clinical ECG, Apple Watch, Samsung, AliveCor, other

What must NOT be stored: Raw ECG waveform data, precise GPS coordinates (only country-level), date of birth, name or any direct identifier, device serial number, any PQRST morphological features, or any data that could enable re-identification. The raw ECG should be processed entirely on-device, with only derived artistic parameters transmitted to the server.

Preventing the extraction feeling

Trust design must be embedded at every layer. Five evidence-based patterns:

1. **Radical transparency**: Show exactly what data leaves the device (artistic parameters only) and what does not (raw ECG). Research on community-based participatory design consistently identifies transparency as the primary trust driver. Springer
2. **Agency over narrative**: Contributors choose whether their tile is public or private, whether to disclose a condition category, and how their contribution appears — mirroring the AIDS Quilt's model where each maker controls representation.
3. **Bidirectional value**: The contribution must immediately return value to the contributor — seeing their unique art tile, seeing the counter increment, receiving milestone celebrations. The Red Cross model of notifying blood donors when their blood reaches a hospital is the gold standard for closing the feedback loop. American Red Cross
4. **No data commercialization**: Contributed rhythms should never be sold, shared with third parties, or used for any purpose beyond the community feature. PatientsLikeMe's erosion

of trust after data commercialization is a direct cautionary precedent. Cambridge Core

5. **Contribution ≠ condition disclosure**: A user can contribute a rhythm tile without revealing they have any specific condition. The art tile's visual characteristics are abstracted from the source signal — a tile from an AF patient looks different from a tile from a sinus rhythm user, but the difference is aesthetic, not diagnostic.

Narrative framing and milestone storytelling

The language of shared rhythm

Clinical language alienates. Research shows patient education materials for cardiovascular conditions average an **11th-grade reading level** — far above the recommended 6th–8th grade. PubMed Central The AMA Manual of Style directs: "Avoid labeling people with their diseases." PubMed Central For OneRhythm, the language framework should center empowerment and collective identity:

Clinical frame (avoid)	Community frame (use)
"ECG data collected"	"Heart rhythms shared"
"Patient contributions"	"Community rhythms"
"Aggregate monitoring data"	"Our shared rhythm journey"
"Data points recorded"	"Moments of rhythm contributed"
"Arrhythmia metrics"	"The rhythm we share"

The word **"rhythm"** is the anchor. It evokes music, life, movement, and pattern — not clinical surveillance. "Shared" creates belonging; "collected" implies extraction. The contributor is an active participant, not a passive data source.

A milestone ladder from the English Channel to the Moon

Distance milestones should follow a narrative arc from intimate to planetary, each paired with a story that makes the number feel human. The following ladder uses verified distances:

Distance	Milestone narrative	Contributions needed (standard)	Contributions needed (wearable)
34 km	"Our shared rhythm just crossed the English Channel"	136,000	45,333
340 km	"Together, our rhythm stretches from London to Paris"	1,360,000	453,333
6,437 km	"Our rhythm flows the entire length of the Amazon River"	25,748,000	8,582,667
7,100 km	"Our shared rhythm connects Tampa to London"	28,400,000	9,466,667
10,849 km	"Our hearts reach from New York to Tokyo"	43,396,000	14,465,333
13,138 km	"Every AF patient on Earth, one rhythm each"	52,550,000	17,516,667
21,196 km	"Our rhythm spans the entire Great Wall of China"	84,784,000	28,261,333
40,075 km	"Together, we have circled the Earth"	160,300,068	53,433,356
384,400 km	"Our shared rhythm has reached the Moon"	1,537,600,000	512,533,333

Movember's success demonstrates that **1:1 symbolic ratios** drive emotional resonance — their 60 km goal maps to 60 men lost to suicide every hour. Awareness Days Calm OneRhythm should frame each contribution similarly: "Every rhythm you share adds 25 centimeters to our collective journey." The American Cancer Society's Relay for Life shows that named ceremonies at transitions (Survivor Lap, Luminaria Ceremony, Fight Back Ceremony) transform metrics into emotional moments. Wikipedia Folding@home's superlative comparisons ("faster than the top 500 supercomputers combined") demonstrate that collective achievement framing generates awe and sustained participation. HotHardware +2

Geographic storytelling that resonates

The most effective distance metaphors share three qualities: **cultural familiarity** (the English Channel, "around the world"), **imaginability** (people can picture Tampa to London), and **progressive escalation** (crossing the Channel feels achievable; reaching the Moon feels

audacious). Charity Miles' motto "#EveryMileMatters" (Charity Miles App) and Movember's team-specific geographic mapping (one team framed 3,000 km as "our most northwest facility to our most southeast") (Movember) show that local personalization amplifies impact. OneRhythm should complement global milestones with personalized ones: "Contributors from the UK have stretched their shared rhythm from Edinburgh to London."

Conclusion: design principles for the Shared Rhythm Distance feature

This feature sits at the intersection of verified clinical need, sound privacy engineering, and established behavioral science. The epidemiological foundation is solid: **52.55 million AF patients** (PubMed Central) and an estimated **100+ million people** with any cardiac arrhythmia globally, with lifetime risk approaching 1 in 3. (ESC) The math is narratively powerful: the AF cohort alone, contributing wearable-length strips, generates **98.3% of an Earth circumnavigation** — close enough to make the "circle the Earth" goal feel achievable rather than abstract.

Three design principles should guide implementation. First, **the artistic transformation is the privacy architecture** — by extracting only non-biometric features on-device and generating one-way visual mappings, the art tile becomes both the product and the anonymization layer, eliminating the need to store or transmit raw ECG waveforms. Second, **the counter must feel like a collective creation, not a data collection** — the IKEA effect, (Wikipedia) (ResearchGate) goal-gradient psychology, and AIDS Quilt precedent all confirm that when people build something together and retain agency over their contributions, they develop ownership and belonging rather than feeling extracted. Third, **milestones should be narrated, not merely triggered** — each threshold is an opportunity for geographic storytelling, community celebration, and renewed motivation through the endowed progress effect.

The most novel insight from this research: the convergence of wearable ECG devices on a 30-second recording standard means that approximately **53.4 million wearable contributions** — remarkably close to the global AF population of 52.55 million — would complete one full circumnavigation. This is not a coincidence to be manufactured but a coincidence to be highlighted. The feature's tagline practically writes itself: *Every rhythm shared. Every heartbeat counted. Together, we circle the Earth.*